	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 2

Lab Title: Introduction to Arrays

Group Members: Muhammad Waqas **Reg. No:** 240756

Objective: Learn what arrays are and how they work in C++.Practice how to create, use, and solve problems using 1D and 2D arrays.

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Computer Engineering

Data Structures And Algorithms

LAB REPORT : 2

Tittle : Introduction To Arrays

Name : Muhammad Waqas

Roll No: 240756

Submitted To : CE . Engr Maryam

Date : 27 September 2025

Question 1 :

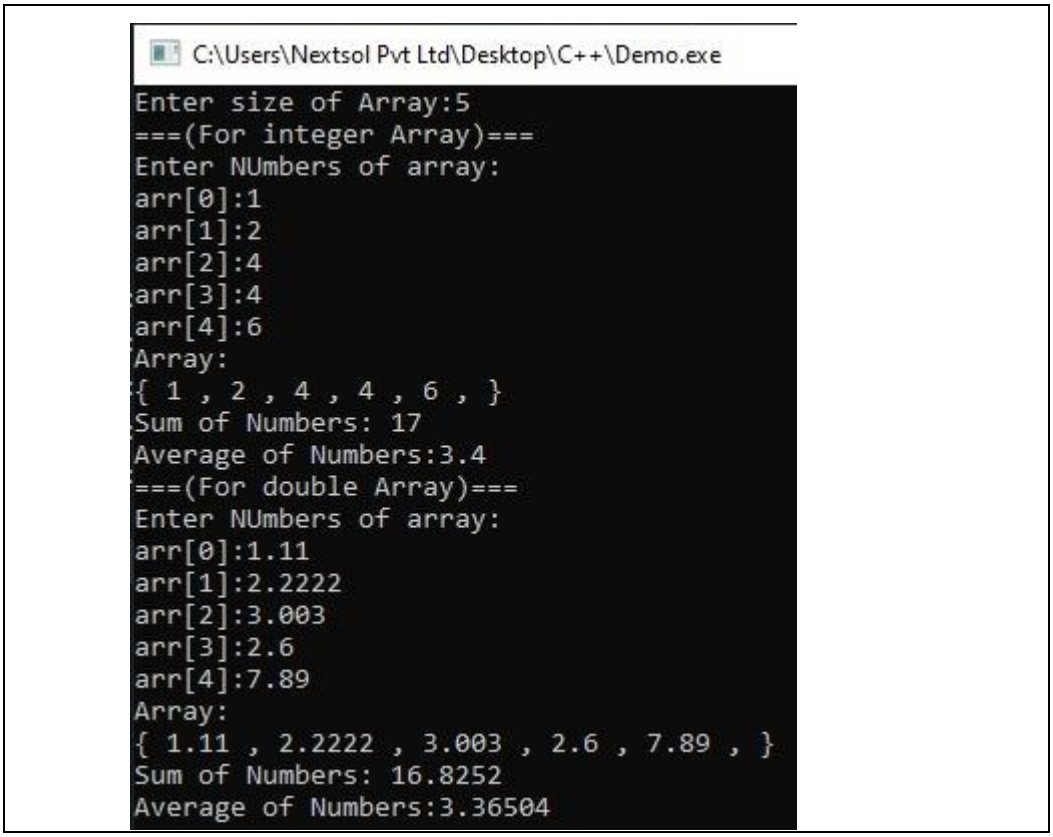
Calculate Sum and Average of N Numbers

With class

most.cpp > Display_Data()

```cpp
#include<iostream>
using namespace std;
template<typename T>
class Most_class{
private:
T* arr;
int n;
public:
Most_class(int n1);
~Most_class();
void Enter_Data();
void Display_Data();
void repeated_var(); };
template<typename T>
Most_class<T>::Most_class(int n1){
n=n1;
arr=new T[n];
for(int i=0;i<n;i++){
arr[i]=0;}}
template<typename T>
Most_class<T>::~Most_class(){
delete[] arr;
cout<<"Destructor\n";}
template<typename T>
void Most_class<T>::Enter_Data(){
cout<<"Enter array elements:\n";
for(int i=0;i<n;i++){
cout<<"a["<<i<<"]: ";cin>>arr[i];}}
template<typename T>
void Most_class<T>::Display_Data(){
cout<<"{";
for(int i=0;i<n;i++){
 cout<<"Array:\n";
cout<<"{ ";
for(int i=0;i<n;i++){
cout<<arr[i]<<" , "; }
 cout<<"}";}
template< typename T>
void Array<T>::average(){
double sum=0;
for(int i=0;i<n;i++){
sum+=arr[i];    }
double avg=static_cast<double>(sum)/n;
cout<<"\nSum of Numbers: "<<sum<<endl;
cout<<"Average of Numbers:"<<avg<<endl;}
int main(){
int size1;
cout<<"Enter size of Array:";
cin>>size1;
 Array<int> s(size1); //For int array
Array<double> f1(size1); //For double arr

cout<<"===(For integer Array)===\n";
s.EnterData();
s.display_data();
s.average();
cout<<"===(For double Array)===\n";
 f1.EnterData();
 f1.display_data();
 f1.average();
system("pause");
cout<<"End";
 return 0;}
```


C:\Users\Nextsol Pvt Ltd\Desktop\C++\Demo.exe
Enter size of Array:5
===(For integer Array)===
Enter NUmbers of array:
arr[0]:1
arr[1]:2
arr[2]:4
arr[3]:4
arr[4]:6
Array:
{ 1 , 2 , 4 , 4 , 6 , }
Sum of Numbers: 17
Average of Numbers:3.4
===(For double Array)===
Enter NUmbers of array:
arr[0]:1.11
arr[1]:2.2222
arr[2]:3.003
arr[3]:2.6
arr[4]:7.89
Array:
{ 1.11 , 2.2222 , 3.003 , 2.6 , 7.89 , }
Sum of Numbers: 16.8252
Average of Numbers:3.36504

Question 2 :

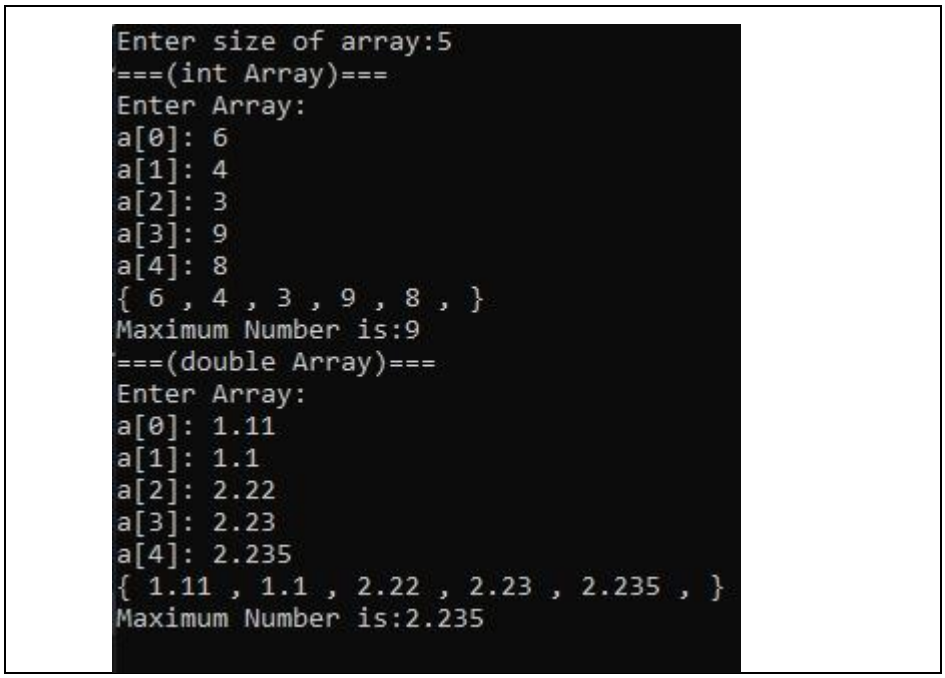
Write a C++ program to find the largest element of a given array of integers (using a class, no vector).

most.cpp > Display_Data()

```cpp
#include<iostream>
using namespace std;
template<typename T>
class Most_class{
private:
T* arr;
int n;
public:
Most_class(int n1);
~Most_class();
void Enter_Data();
void Display_Data();
void repeated_var(); };
template<typename T>
Most_class<T>::Most_class(int n1){
n=n1;
arr=new T[n];
for(int i=0;i<n;i++){
arr[i]=0;}}
template<typename T>
Most_class<T>::~Most_class(){
delete[] arr;
cout<<"Destructor\n";}
template<typename T>
void Most_class<T>::Enter_Data(){
cout<<"Enter array elements:\n";
for(int i=0;i<n;i++){
cout<<"a["<<i<<"]: ";cin>>arr[i];}}
template<typename T>
void Most_class<T>::Display_Data(){
cout<<"{";
for(int i=0;i<n;i++){
```

```cpp
void Max_class<T>::display_data(){
cout<<"{ ";
for(int i=0;i<n;i++){
cout<<arr[i]<<" , ";}
cout<<"}\n";}
template< typename T>
void Max_class<T>::max_number(){
T maxnum=arr[0];
for(int i=0;i<n;i++){
if(maxnum<arr[i]){
maxnum=arr[i];}}
cout<<"Maximum Number is:"<<maxnum<<endl;}
int main(){
int size1;
cout<<"Enter size of array:";
cin>>size1;
Max_class<int> a1(size1);
Max_class<double> b1(size1);
cout<<"===(int Array)===\n";
a1.Enter_data();
a1.display_data();
a1.max_number();
cout<<"===(double Array)===\n";
b1.Enter_data();
b1.display_data();
b1.max_number();
return 0;}
```

Output:

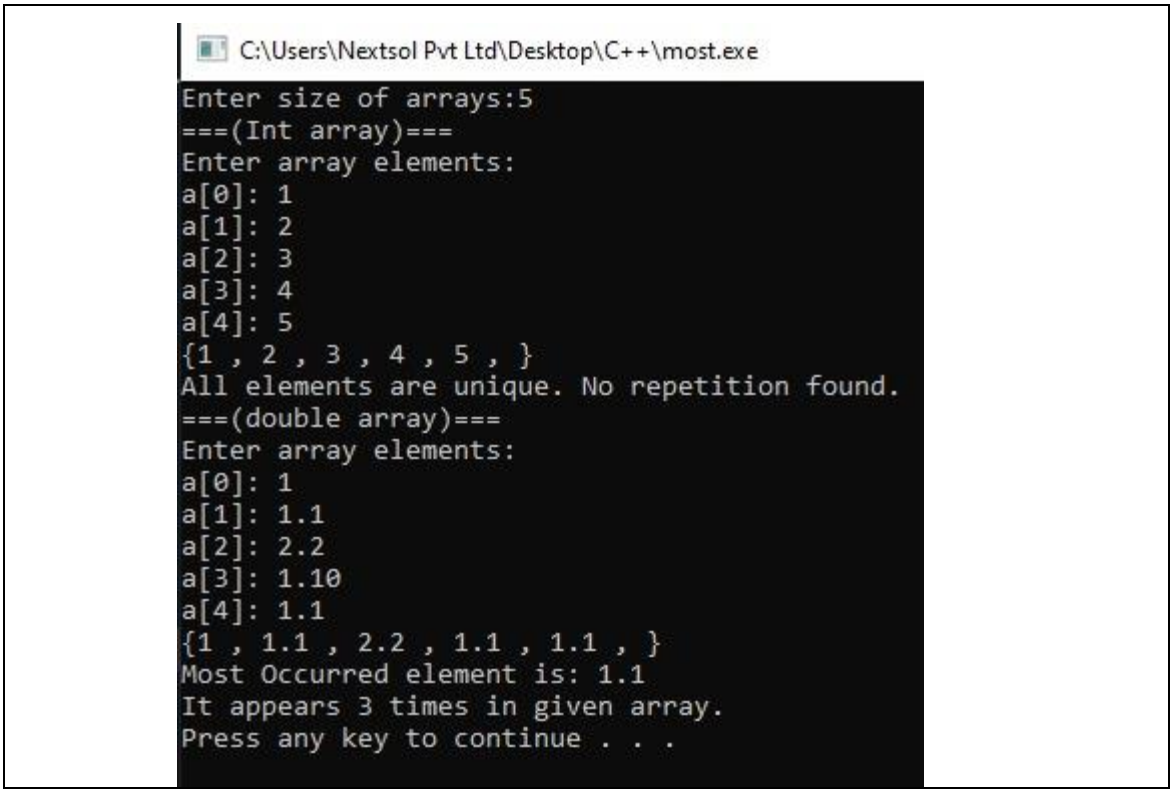

Enter size of array:5
===(int Array)===
Enter Array:
a[0]: 6
a[1]: 4
a[2]: 3
a[3]: 9
a[4]: 8
{ 6 , 4 , 3 , 9 , 8 , }
Maximum Number is:9
===(double Array)===
Enter Array:
a[0]: 1.11
a[1]: 1.1
a[2]: 2.22
a[3]: 2.23
a[4]: 2.235
{ 1.11 , 1.1 , 2.22 , 2.23 , 2.235 , }
Maximum Number is:2.235

Question 3 :

Write a C++ program to find the most occurring element in an array of integers (using a class, no vector).

most.cpp > Display_Data()

```cpp
#include<iostream>
using namespace std;
template<typename T>
class Most_class{
private:
T* arr;
int n;
public:
Most_class(int n1);
~Most_class();
void Enter_Data();
void Display_Data();
void repeated_var(); };
template<typename T>
Most_class<T>::Most_class(int n1){
n=n1;
arr=new T[n];
for(int i=0;i<n;i++){
arr[i]=0;}}
template<typename T>
Most_class<T>::~Most_class(){
delete[] arr;
cout<<"Destructor\n";}
template<typename T>
void Most_class<T>::Enter_Data(){
cout<<"Enter array elements:\n";
for(int i=0;i<n;i++){
cout<<"a["<<i<<"]: ";cin>>arr[i];}}
template<typename T>
void Most_class<T>::Display_Data(){
cout<<"{";
for(int i=0;i<n;i++){
for(int i=0;i<n;i++){
cout<<arr[i]<<" , ";}
cout<<"}\n";}
template<typename T>
void Most_class<T>::repeated_var(){
int maxcount = 0;
T most_var = arr[0];
for(int i=0;i<n;i++){ int count=0;
for(int j=0;j<n;j++){
if(arr[i]==arr[j]){count++;}}
if(count>maxcount){
maxcount=count;
most_var=arr[i];}}
if(maxcount == 1) {
cout << "All elements are unique.";
} else {
cout << "Most Occurred element is: "
<< most_var << endl;
cout << "It appears " << maxcount
<< " times in given array.\n";}}
int main(){ int size;
cout<<"Enter size of arrays:";cin>>size;
Most_class<int> m1(size);
Most_class<double> m2(size);
cout<<"===(Int array)===\n";
m1.Enter_Data(); m1.Display_Data();
m1.repeated_var();
cout<<"===(double array)===\n";
m2.Enter_Data();m2.Display_Data();
m2.repeated_var();
system("pause");return 0;}
```


C:\Users\Nextsol Pvt Ltd\Desktop\C++\most.exe
Enter size of arrays:5
===(Int array)===
Enter array elements:
a[0]: 1
a[1]: 2
a[2]: 3
a[3]: 4
a[4]: 5
{1 , 2 , 3 , 4 , 5 , }
All elements are unique. No repetition found.
===(double array)===
Enter array elements:
a[0]: 1
a[1]: 1.1
a[2]: 2.2
a[3]: 1.10
a[4]: 1.1
{1 , 1.1 , 2.2 , 1.1 , 1.1 , }
Most Occurred element is: 1.1
It appears 3 times in given array.
Press any key to continue . . .

Conclusion:

- Arrays store multiple values in a single variable using a fixed size.
- Pointers can be used to access and manipulate elements in an array.
- The name of an array acts like a pointer to its first element.
- Using pointers with arrays helps in dynamic memory handling and efficient looping.
- Changes made using pointers directly affect the original array elements.
- Understanding arrays with pointers is important for memory management in DSA.